



COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

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Unit – 1 Computer Networks and the Internet

1.1 Introduction to Computer Networks

1.2 What is the Internet?

- A nuts-and-bolts and Services description, Protocol

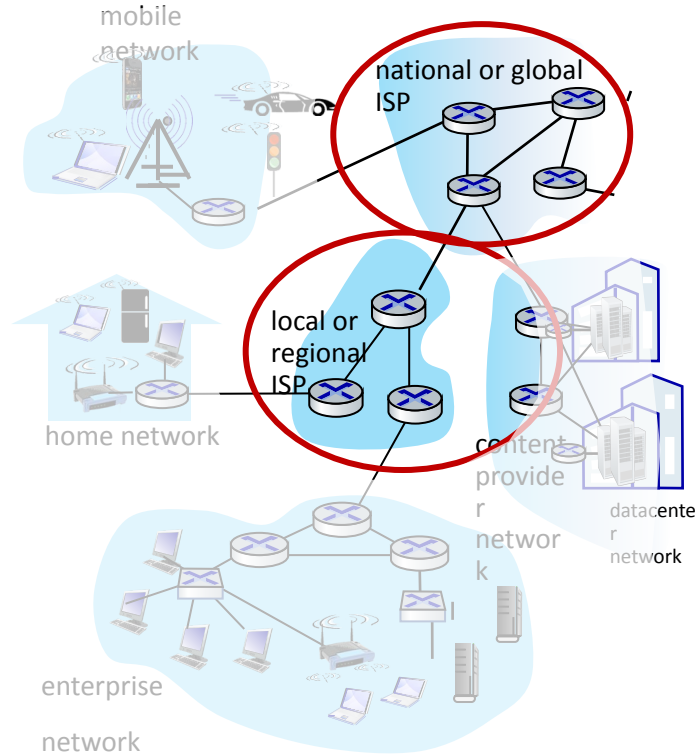
1.3 Network edge

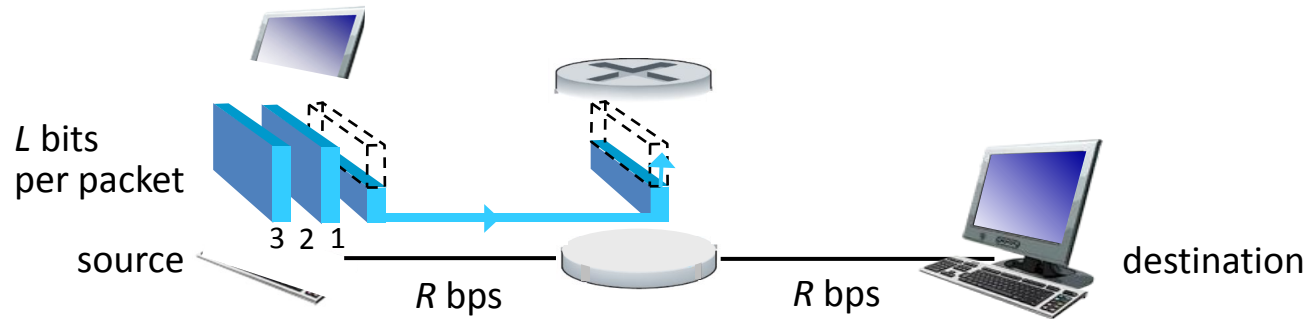
- End systems, Access networks, Physical media

1.4 Network core

- Packet switching, Circuit switching, Network structure

- mesh of interconnected routers
- **packet-switching**: hosts break application-layer messages into *packets*
 - forward packets from one router to the next, across links on path from source to destination
 - each packet transmitted at full link capacity

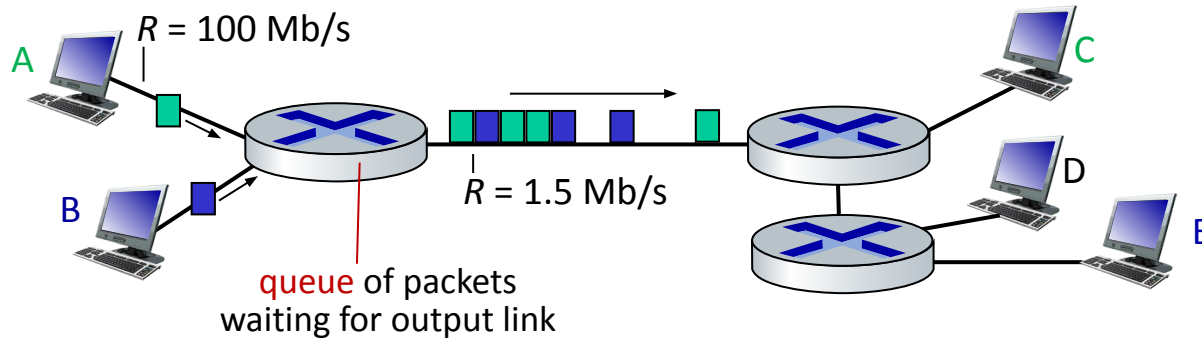




- **Transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **Store and forward:** entire packet must arrive at router before it can be transmitted on next link
- **End-end delay:** $2L/R$ (above), assuming zero propagation delay (more on delay shortly)

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- one-hop transmission delay = 0.1 msec

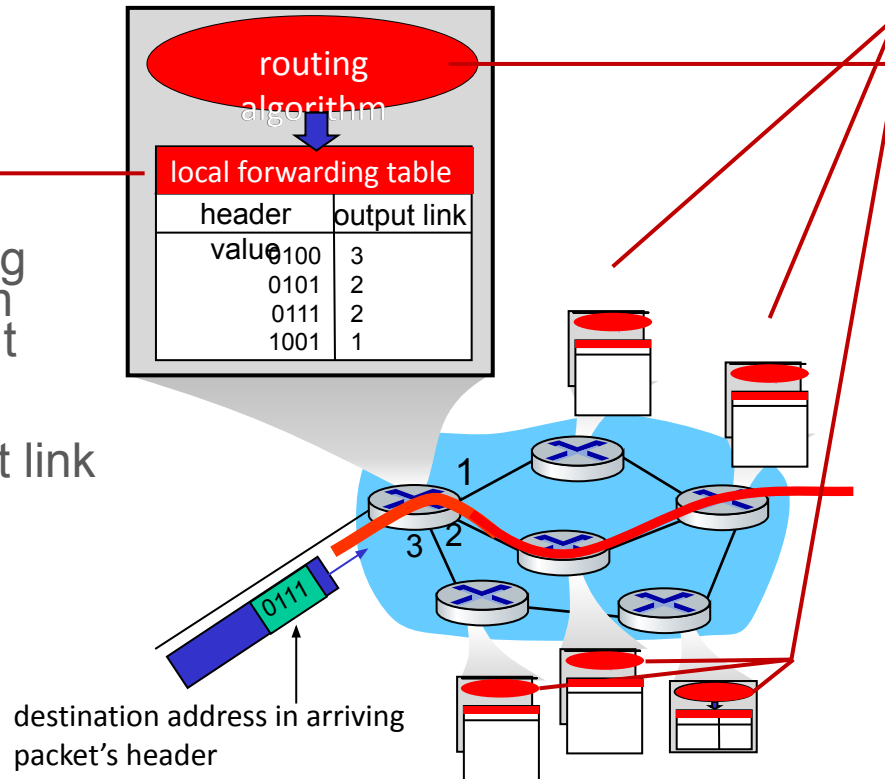


Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for a period of time:

- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

Forwarding:

- **local** action: move arriving packets from router's input link to appropriate router output link



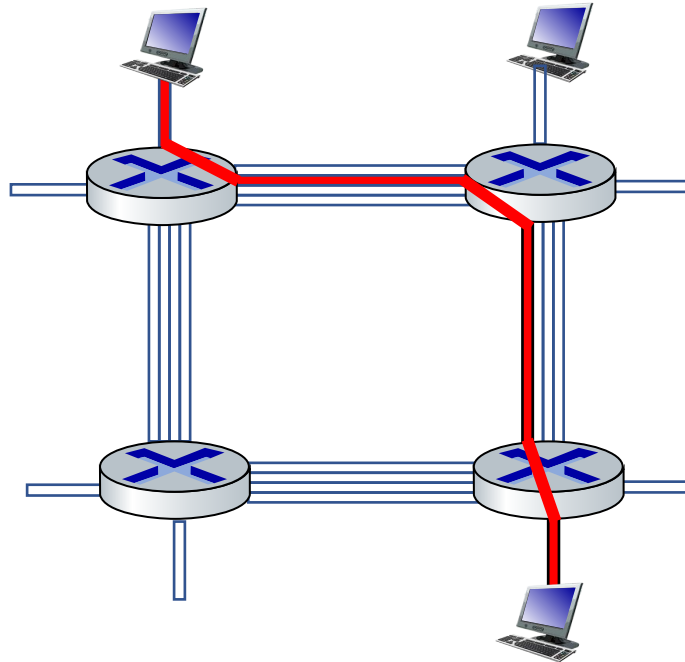
Routing:

- **global** action: determine source-destination paths taken by packets
- routing algorithms

Network Core: Circuit Switching

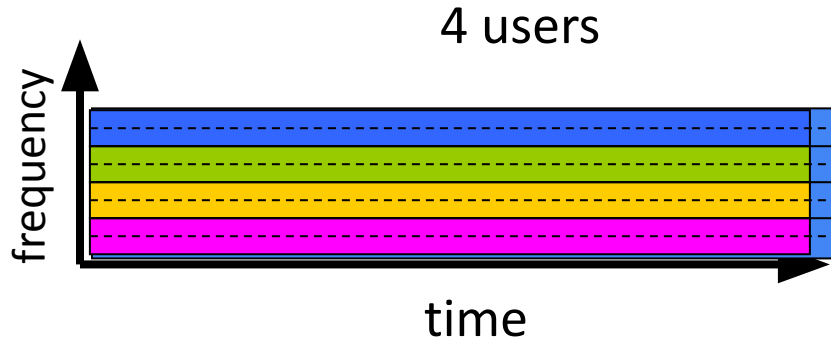
end-end resources allocated to, reserved for “call” between source and destination (eg: telephone)

- in diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (**no sharing**)
- commonly used in traditional telephone networks



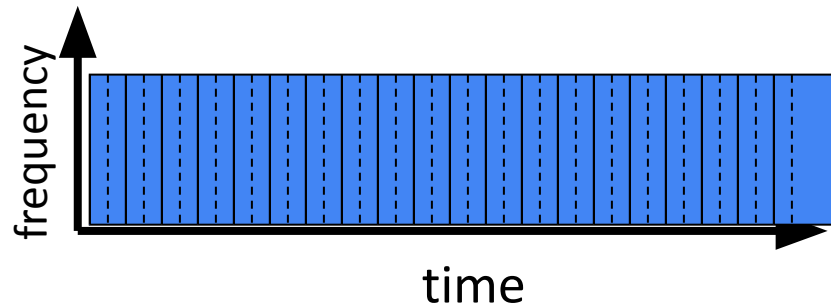
Frequency Division Multiplexing (FDM)

- optical, electromagnetic frequencies divided into (narrow) frequency bands
- each call allocated its own band, can transmit at max rate of that narrow band



Time Division Multiplexing (TDM)

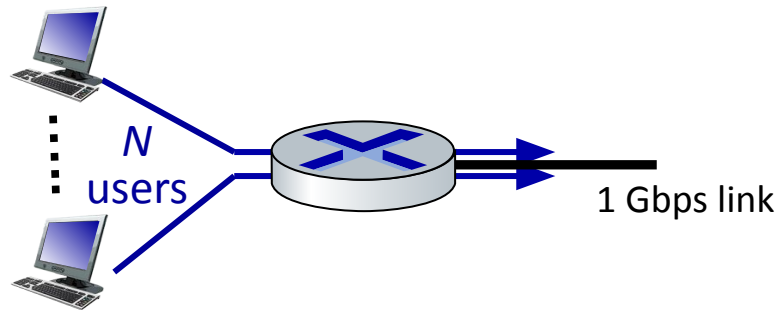
- time divided into frames -> slots
- each call allocated periodic slot(s), can transmit at maximum rate of (wider) frequency band, but only during its time slot(s)



packet switching allows more users to use network!

Example:

- 1 Gb/s link
- each user:
 - 100 Mb/s when “active”
 - active 10% of time



- *circuit-switching*: 10 users
- *packet switching*: with 35 users,
 - probability > 10 active users at same time is less than .0004 *
 - 10 or few active users, probability 0.9996

Q: how did we get value 0.0004?

Q: what happens if > 35 users ?

Is packet switching a “slam dunk winner”?

- great for “bursty” data – sometimes has data to send, but at other times not
 - resource sharing
 - simpler, no call setup
- **excessive congestion possible:** packet delay and loss due to buffer overflow
 - protocols needed for reliable data transfer, congestion control
- **Q: How to provide circuit-like behavior?**
 - bandwidth guarantees traditionally used for audio/video applications

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet switching)?

- How long does it take to send a file of 640,000 bits (1 byte = 8 bits) from host A to host B over a circuit-switched network?
 - All links are 1.536 Mbps
 - Each link uses TDM with 24 slots/sec
 - 500 msec to establish end-to-end circuit

Let's work it out!

Solution:

- Each circuit has a transmission rate of $(1.536 \text{ Mbps})/24 = 64 \text{ kbps}$
- It takes $(640,000 \text{ bits})/(64 \text{ kbps}) = 10 \text{ seconds}$ to transmit the file
- To this 10 seconds we add the circuit establishment time, giving 10.5 seconds to send the file



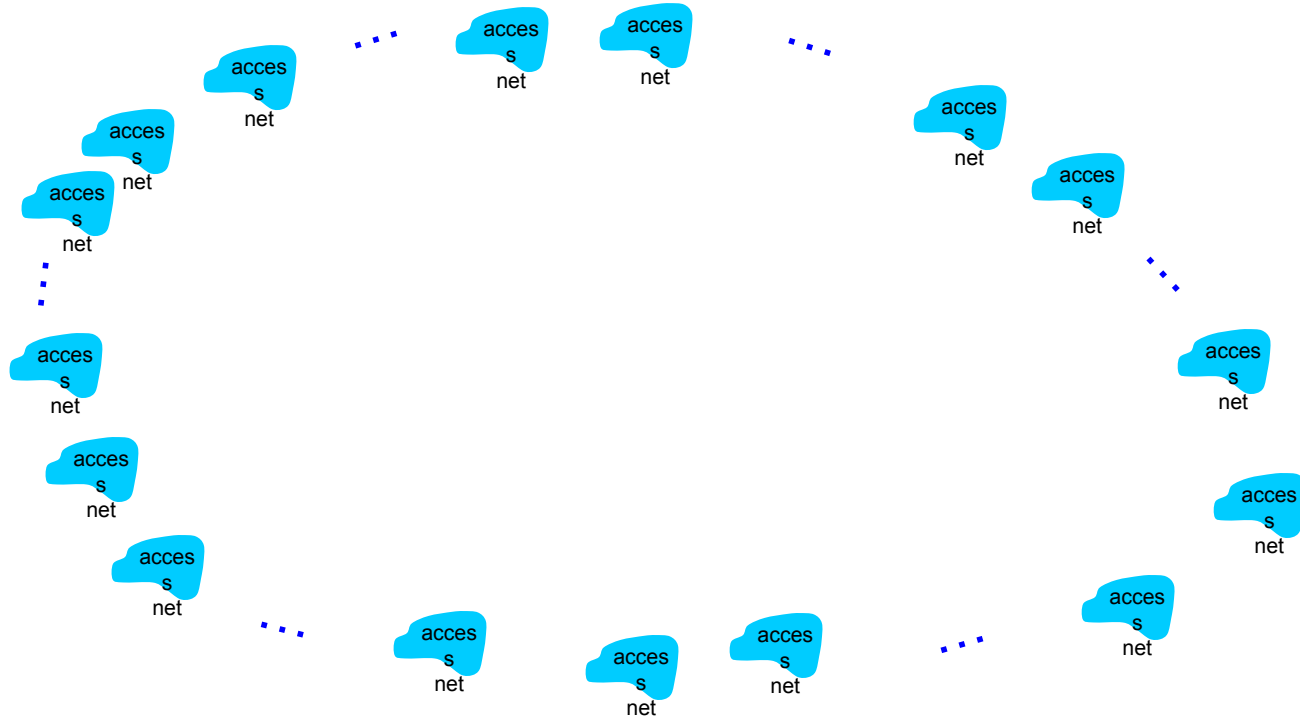
- | | |
|--|----------------------------------|
| • Connectionless | • Connection oriented |
| • Designed for data | • Designed for voice |
| • Flexible | • Inflexible |
| • Out of order, assembled at the dest | • Message received in same order |
| • Forward, Store & Fwd | • FDM & TDM |
| • Network layer | • Physical layer |
| • Bandwidth is saved (dynamic) | • Bandwidth is wasted (fixed) |
| • Transmission of data – Source, routers | • Transmission of data – source |
| • Transmission delay | • Call setup delay |

Internet Structure: a “network of networks”

- ❖ End systems connect to Internet via **access ISPs** (Internet Service Providers)
 - Residential, company and university ISPs
- ❖ Access ISPs in turn must be interconnected.
 - ❖ So that any two hosts can send packets to each other
- ❖ Resulting network of networks is very complex
 - ❖ Evolution was driven by **economics** and **national policies**
- ❖ Let's take a stepwise approach to describe current Internet structure

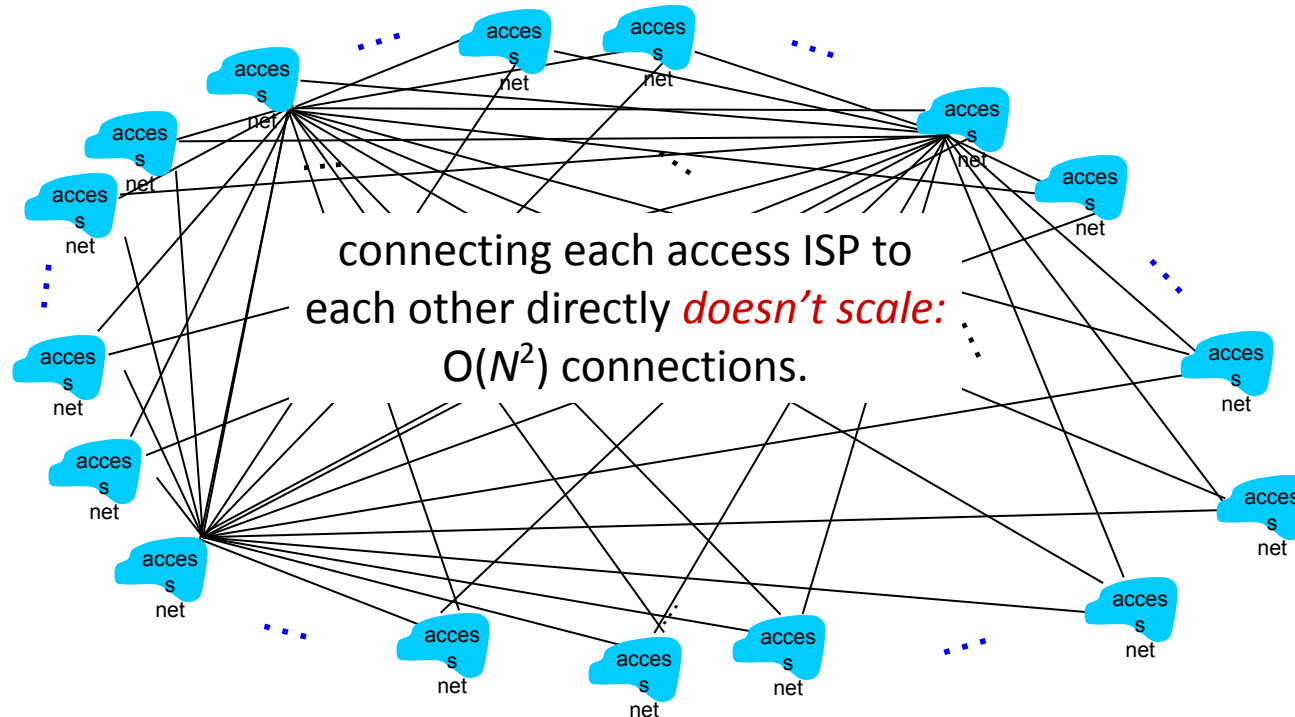
Internet Structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



Internet Structure: a “network of networks”

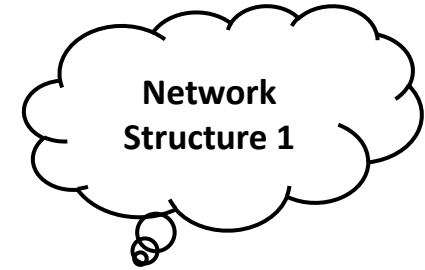
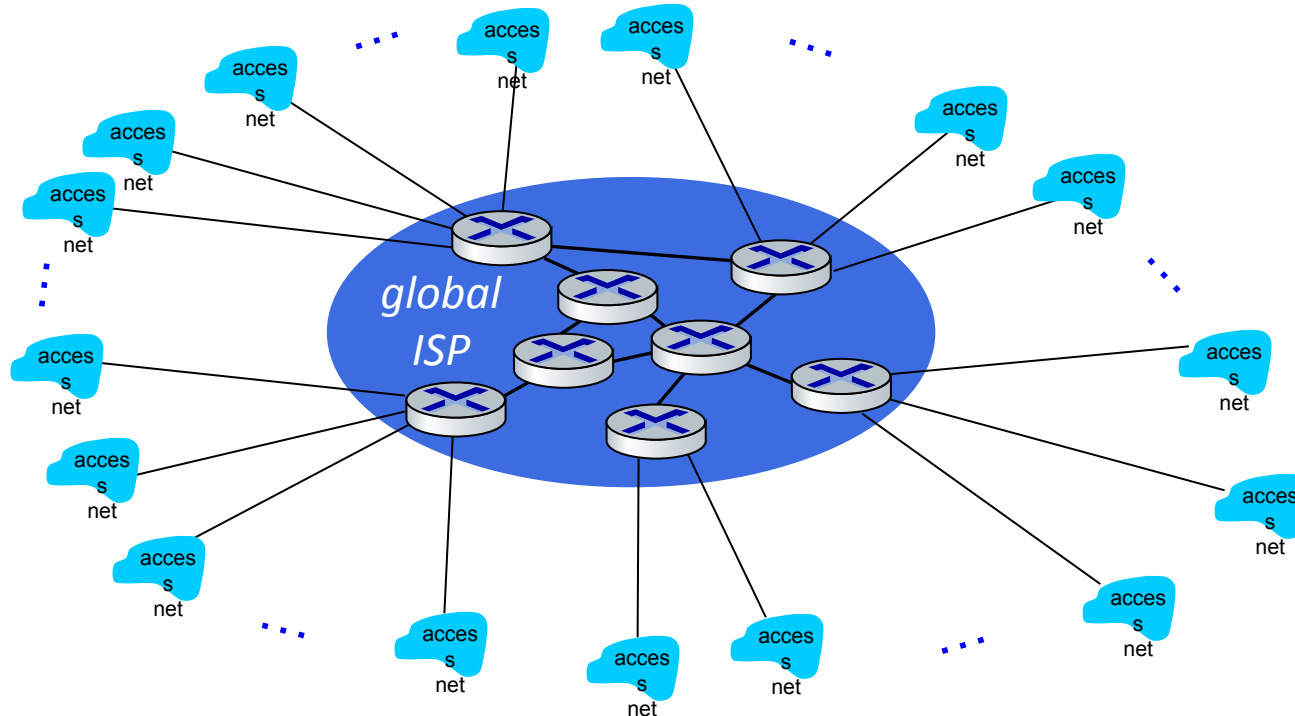
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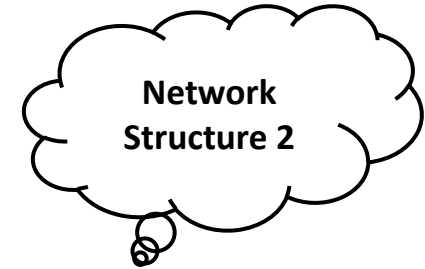
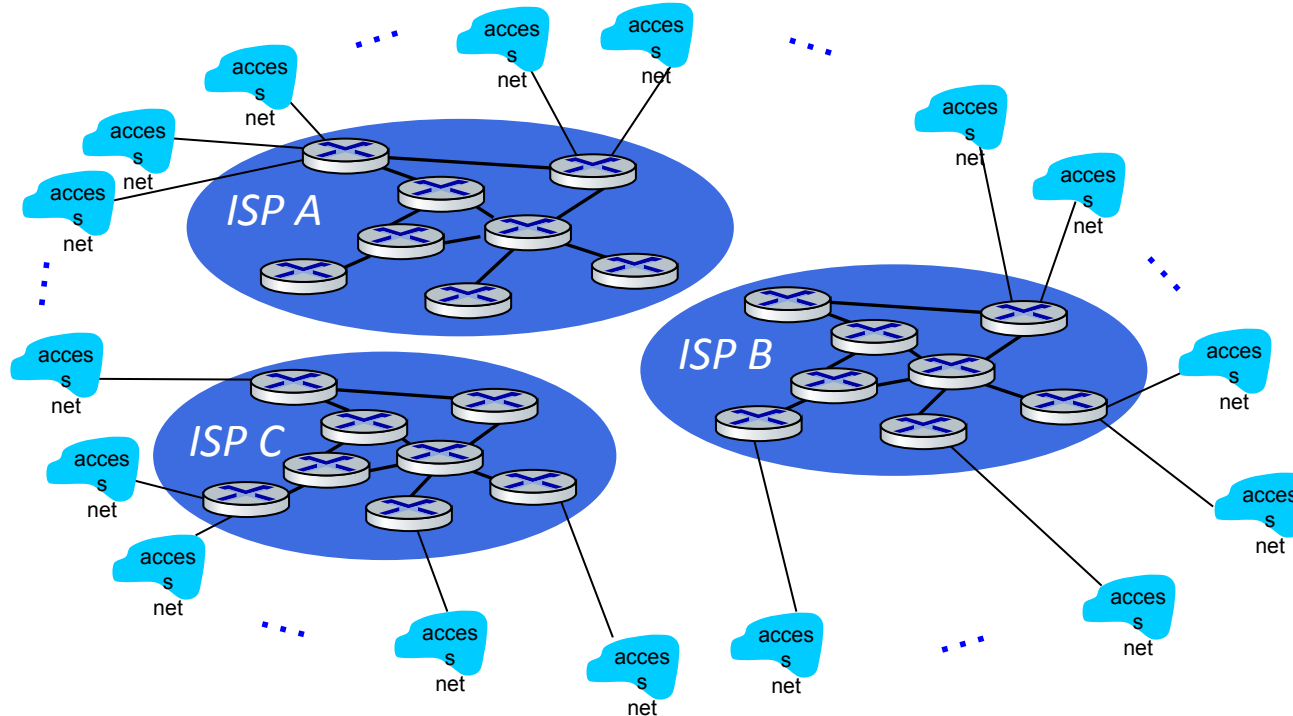
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



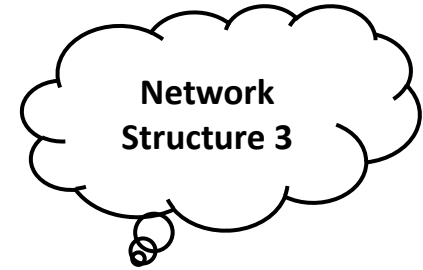
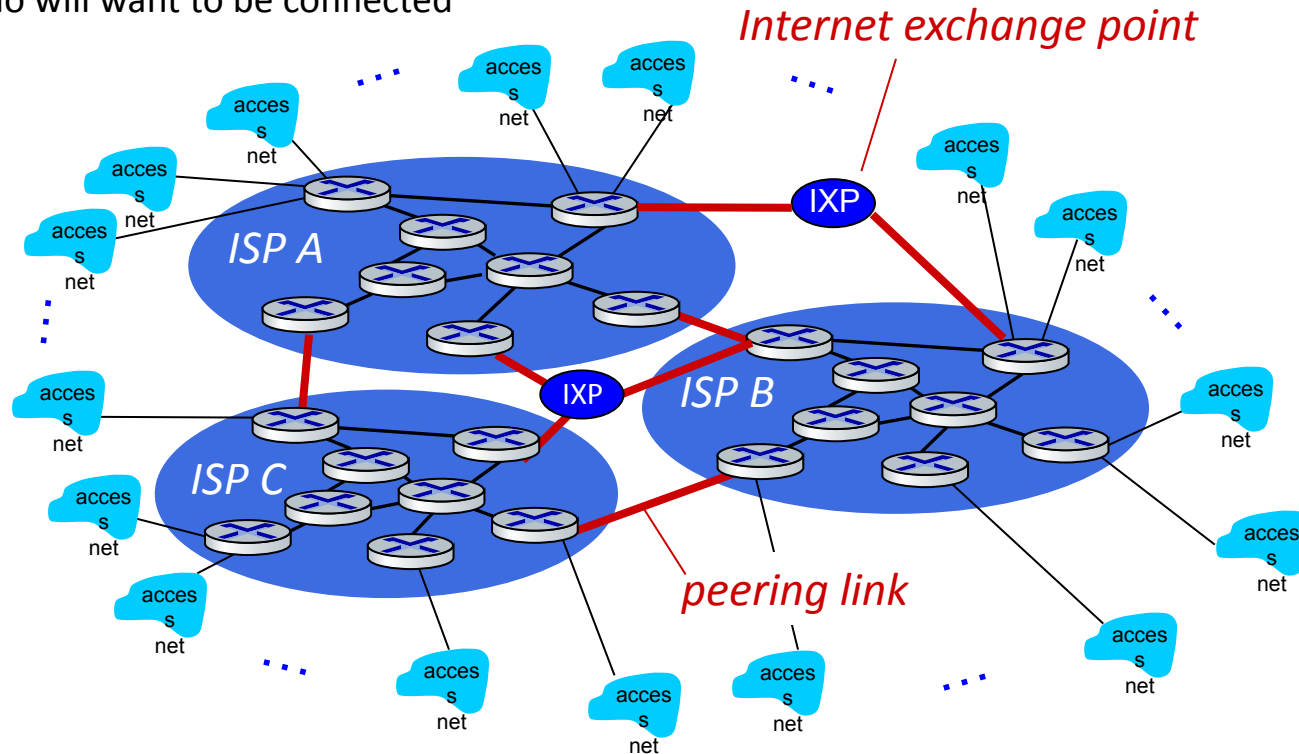
Internet Structure: a “network of networks”

But if one global ISP is viable business, there will be competitors



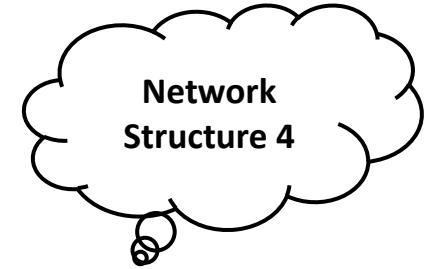
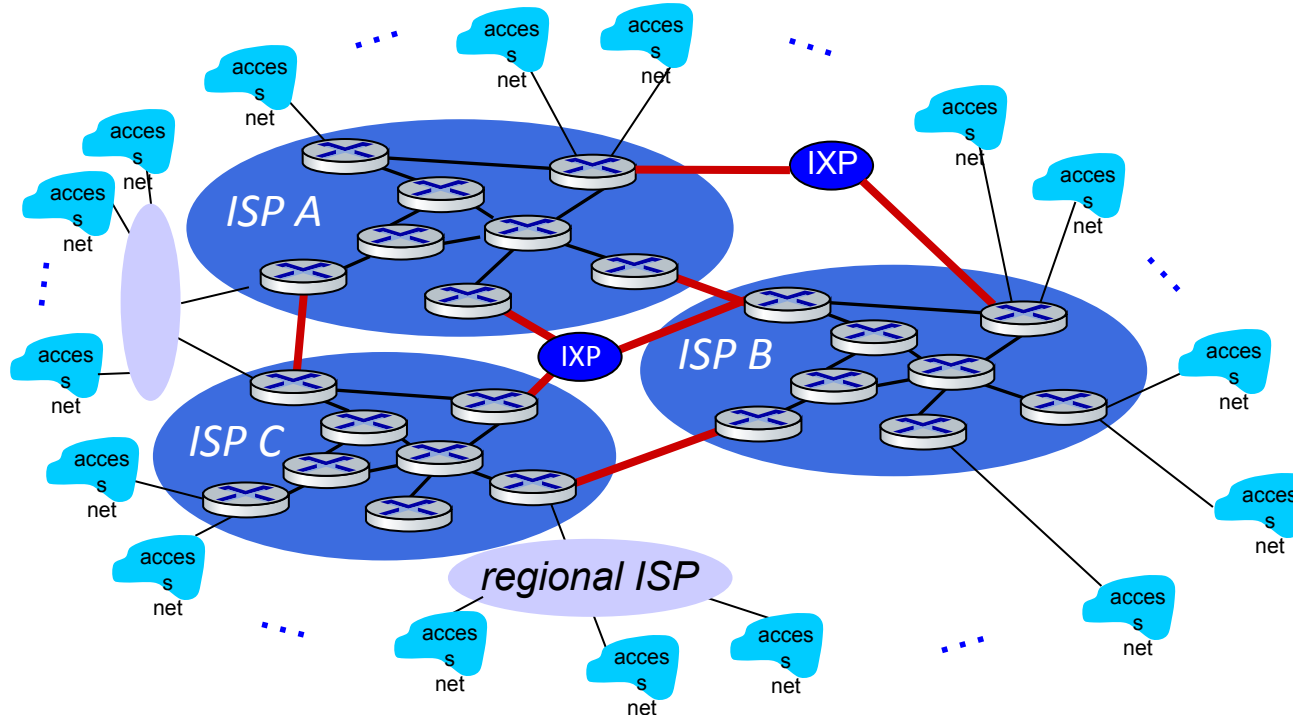
Internet Structure: a “network of networks”

But if one global ISP is viable business, there will be competitors
who will want to be connected



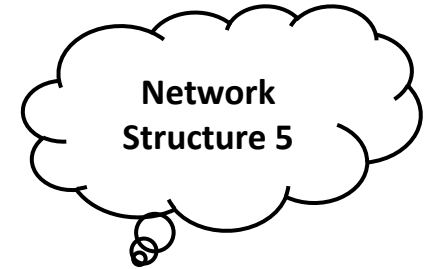
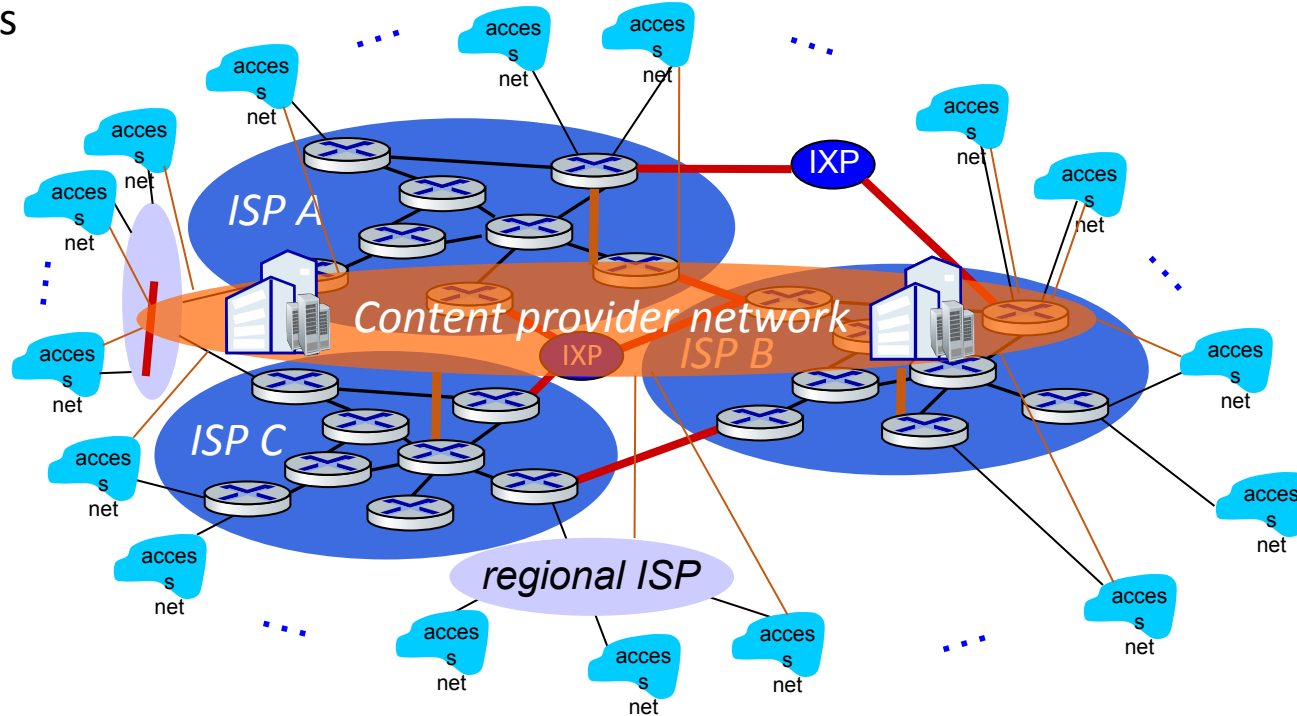
Internet Structure: a “network of networks”

... and regional networks may arise to connect access nets to ISPs

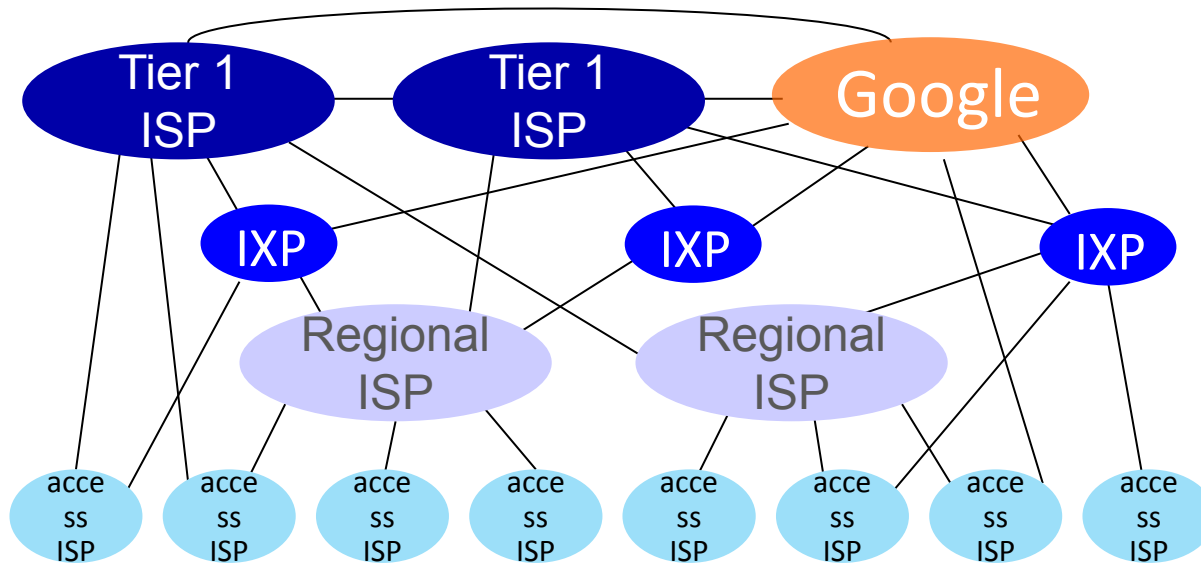


Internet Structure: a “network of networks”

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet Structure: a “network of networks”

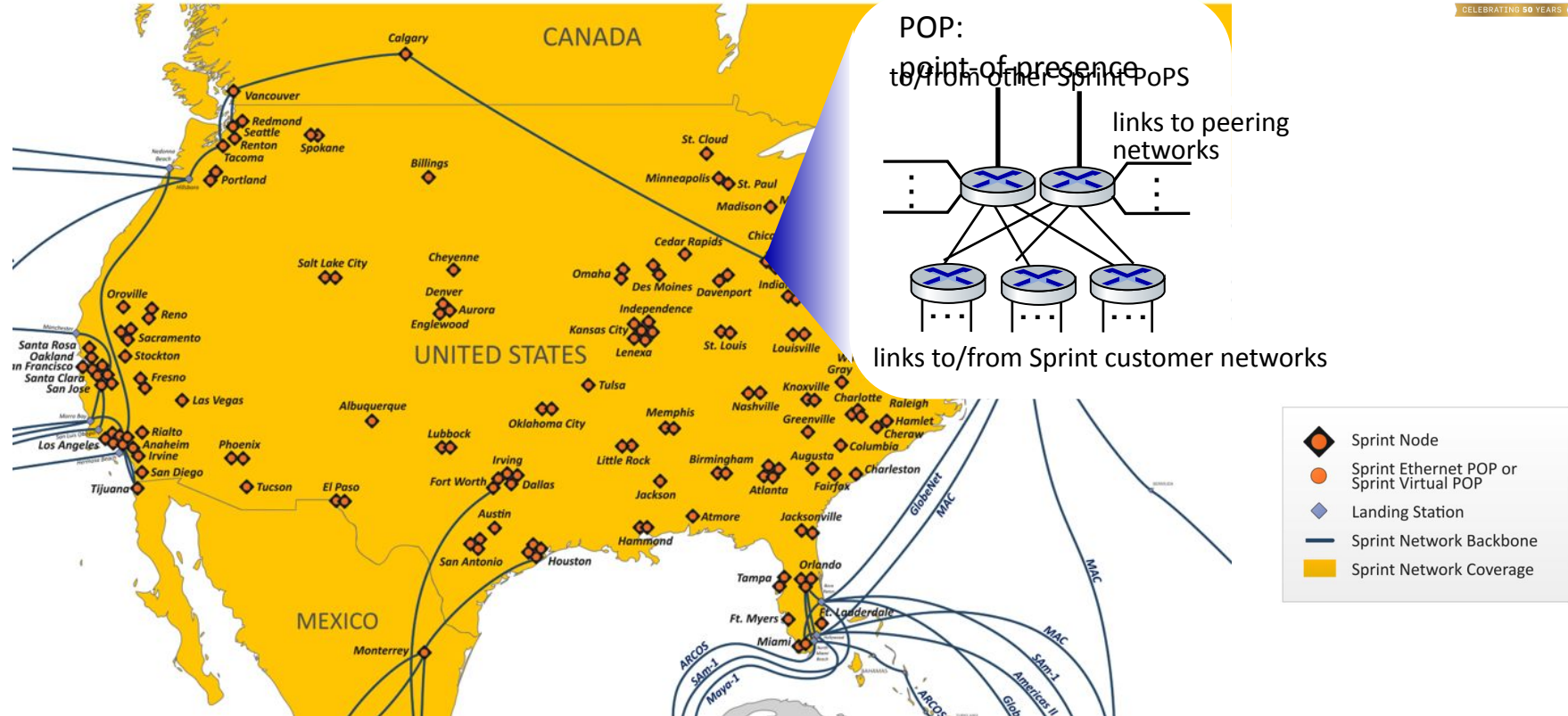


At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

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Network Core: Tier 1 ISP Network Map: Sprint 2019





THANK YOU

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